# Part C — AI Usage Report

**DSC 190/291 — Assignment 4 Student: Zeyu Bian**

## 1. Where I used AI

I used Claude Code in four roles on this assignment:

- **Part A.1 (VC dimension via supports) — proof structure.** I asked AI to confirm the cleanest way to pass from $\Gamma_{\mathcal{H}_k}(n) \le (e^2 dn/k^2)^k$ to the $O(k \log(ed/k))$ VC bound — specifically, which form of the "if $m \le k \log(am/k)$ then $m = O(k \log a)$" lemma to cite. I then verified the constants by hand.
- **Part A.2 (SRM oracle inequality) — bookkeeping of the three statistical costs.** AI helped me lay out the table separating *support cost* $k \log(ed/k)$, *level cost* $\log(1/p_k)$, and *confidence cost* $\log(1/\delta)$, and made me state the class-level SRM theorem cleanly before plugging things in. The actual instantiation (each $p_k\delta$ confidence parameter, the $2 \cdot \text{pen}$ factor) I wrote and checked myself.
- **Part A.4 (Validation oracle bound) — independence argument.** I asked AI to be a hostile reader on my proof that conditioning on $S_1$ makes the candidate list $\{w_1, \dots, w_d\}$ data-independent of $S_2$. This caught the place where I had originally tried to use a single uniform-convergence bound on $\mathcal{H}$ instead of the much sharper finite-class Hoeffding-with-union-bound on $d$ fixed candidates.
- **Part B (PAC-Bayes for thresholds) — example construction.** For B.3 I sketched the $(N = 20, \tau = 11, S = ((1, 0)))$ example by hand, then asked AI to double-check the arithmetic $\sum_{t=2}^{21} |t - 11| = 100$ and the resulting $L_{\mathcal{D}}(Q_V) = 0.25$.

## 2. AI suggestions I accepted (and why)

- **The product form $\binom{d}{k}(en/k)^k \le (e^2 dn/k^2)^k$ for A.1.** AI suggested writing the growth-function bound in this "two factors of the same shape" form so the log-linear fixed-point lemma applies immediately. I accepted because it cleans up the algebra and produces the bound with the right constants.
- **Confidence rescaling $\delta \to p_k\delta$ in A.2.** AI's framing — "each level pays $\log(1/p_k)$ because the union bound assigns confidence $p_k\delta$ to level $k$" — is exactly the right way to explain the level cost in one sentence. I accepted and put it in the interpretation paragraph.
- **Validation comparison table in A.4.** AI suggested a four-row table comparing penalty-SRM and validation (sample used to fit; sample used to choose; support cost; level cost). I accepted because it makes the trade-off explicit and answers the "what is paid for the separation?" question directly.
- **Distinguish certificate vs learning in B.4.** AI insisted I add a final paragraph separating "PAC-Bayes lower-bounds the certificate" from "this is not a sample complexity lower bound." This is the actual point of the problem; without it the conclusion reads as a paradox.

## 3. AI suggestions I rejected or substantially modified

- **A.1 lower-bound claim.** AI's first draft asserted $\text{VCdim}(\mathcal{H}_k) \ge k \log(d/k)$ without proof and tried to use it as a "matching lower bound." This is true but nontrivial; I demoted the claim to a sanity-check observation that $\text{VCdim}(\mathcal{H}_k) \ge k$ (which is immediate from $\mathcal{H}_I \subseteq \mathcal{H}_k$) and dropped the unproved matching claim.

- **A.4 single-step union bound.** AI's first attempt at the validation bound tried to run *one* uniform-convergence statement over $\bigcup_k \mathcal{H}_k$ on $S_2$. This is wrong: $S_2$ only sees the *finite* list $w_1, \ldots, w_d$, not all of $\mathcal{H}$; using the full uniform-convergence bound would gives a worse rate ($\sqrt{d/n}$ via VC) instead of the correct $\sqrt{\log d/n}$. I rewrote to make the conditioning on $S_1$ explicit and use Hoeffding on $d$ fixed candidates.
- **B.3 example.** AI's first proposed counterexample had $|V(S)|$ small and the KL inequality went the wrong way. I rewrote with $S = ((1, 0))$, which leaves $V(S) = \{2, \ldots, N+1\}$ (i.e. all but one threshold), making the KL inequality strict and the risk inequality strict.
- **B.4 "$\tau \in \{1, \ldots, N+1\}$" claim.** AI's first proof of the pigeonhole step used the pigeonhole over all $N+1$ thresholds to get $P(h_\tau) \le 1/(N+1)$, then tried to construct $\mathcal{D}_\tau$ using $x = \tau - 1, x = \tau$. But this construction fails for $\tau = 1$ (no $x = 0$) and $\tau = N+1$ (no $x = N+1$). I rewrote to restrict $\tau \in \{2, \ldots, N\}$ and get the slightly weaker $P(h_\tau) \le 1/(N-1)$, matching the assignment's $\log(N-1)$ bound.

## 4. How I verified correctness

- **A.1 VC bound.** Verified the algebra step-by-step: $\binom{d}{k} \le (ed/k)^k$ (standard, true for $1 \le k \le d$); Sauer–Shelah for $\mathcal{H}_I$ at $n \ge k$; the log-linear inversion lemma applied to $m \le k \log_2(e^2 dm/k^2)$. Cross-checked against Shalev-Shwartz–Ben-David Lemma A.2.
- **A.2 SRM theorem.** Cross-checked the statement of the class-level SRM theorem and the $2 \cdot \mathrm{pen}$ factor against SSBD §7.2. Verified the chain $\varepsilon_k(n, \delta) \to \mathrm{pen}(k, n, \delta) = \varepsilon_k(n, p_k \delta)$ produces the boxed bound by direct substitution.
- **A.3 sample complexity.** Verified $s \log(ed/s) + 2 \log s = O(s \log(ed/s))$ for $s \le d$ so absorbing $\log(1/p_s)$ into the leading term is legitimate. Verified the boundary $s \log(d/s) \le d$ algebraically: equivalent to $\log(d/s) \le d/s$, true for $d/s \ge 1$ (which is given since $s \le d$).
- **A.4 validation bound.** Verified the two-step argument $(\star) + (\star\star)$ gives the stated bound by writing out the two events explicitly and applying the union bound at $\delta/2 + \delta/2 = \delta$. Cross-checked the finite-class step against SSBD §11.2.
- **A.5 runtime.** Verified $(ed/k)^k$ is the upper bound on $\binom{d}{k}$ I used; noted polynomial-in-$d$ regime via $k = O(1)$ and superpolynomial regime via $k = \log d$ by direct exponentiation.
- **B.1 VC=1.** Verified by hand the impossibility of the $(1, 0)$ pattern on $x_1 < x_2$ from the definition $h_t(x) = \mathbf{1}[x \ge t]$.
- **B.2 version space.** Verified $a(S) < t \le b(S)$ from the two consistency conditions $\{y_i = 0 \Rightarrow t > x_i\}$ and $\{y_i = 1 \Rightarrow t \le x_i\}$. Checked the edge cases (no negatives: $a(S) = 0$ gives $V = \{1, \ldots, b(S)\}$; no positives: $b(S) = N+1$ gives $V = \{a(S)+1, \ldots, N+1\}$).
- **B.3 arithmetic.** Computed $\sum_{t=2}^{21} |t - 11| = \sum_{j=1}^{9} j + 0 + \sum_{j=1}^{10} j = 45 + 55 = 100$ by hand; verified $L_{\mathcal{D}}(Q_V) = 100/400 = 0.25$ matches the formula.
- **B.4 failure probability.** Verified $\Pr[A^c] = \Pr[\text{all } (\tau - 1)] + \Pr[\text{all } \tau] = 2 \cdot 2^{-n} = 2^{1-n}$ by enumerating the support of $\mathcal{D}_\tau^n$.

## AI workflow updates (5 most recent changes)

1. **"Pigeonhole over the right set" rule.** AI tends to apply pigeonhole over the full set ($\{1, \ldots, N+1\}$ here), then build a construction that only works for *interior* indices. I now check, after any pigeonhole step, whether my downstream construction has edge cases. This caught the $\tau \in \{1, N+1\}$ issue in B.4.
2. **"Finite-class vs uniform-convergence" prompt.** Before applying uniform convergence over a hypothesis class, I now explicitly ask "is this set fixed independently of the sample

I'm evaluating on?" If yes, prefer Hoeffding+union over the (finite) set; if no, pay the VC complexity. This fixed the A.4 validation proof.

3. **Concrete-example-first protocol for PAC-Bayes/KL gaps.** Whenever AI claims an inequality between two information-theoretic quantities (e.g. "KL of $Q_V$ is much smaller than KL of $\delta$"), I require a worked numerical example before trusting it. This is how the B.3 example got written.
4. **Costs-table convention.** I added a rule that every oracle inequality I write must come with a row-by-row explanation of each term in the complexity penalty (support cost, level cost, confidence cost, validation overhead). This produced the interpretation paragraph in A.2 and the comparison table in A.4.
5. **"Reject unproved matching lower bounds."** AI is willing to assert VCdim $\geq \ldots$ matching the upper bound without proof. I now demand either a proof or a citation; if neither, I demote the claim to a weaker sanity check (in A.1: $\mathcal{H}_I \subseteq \mathcal{H}_k \Rightarrow \text{VCdim} \geq k$).

I did not use AI to fabricate any reference; every cited result (SSBD Lemma A.2, SSBD §7.2, §11.2) is one I located and read.